BIO 203 - BIOTUTORIAL

Spatial Biodiversity & Environmental Modeling of Damselfish *(Pomacentridae)* in the Tañon Strait Region

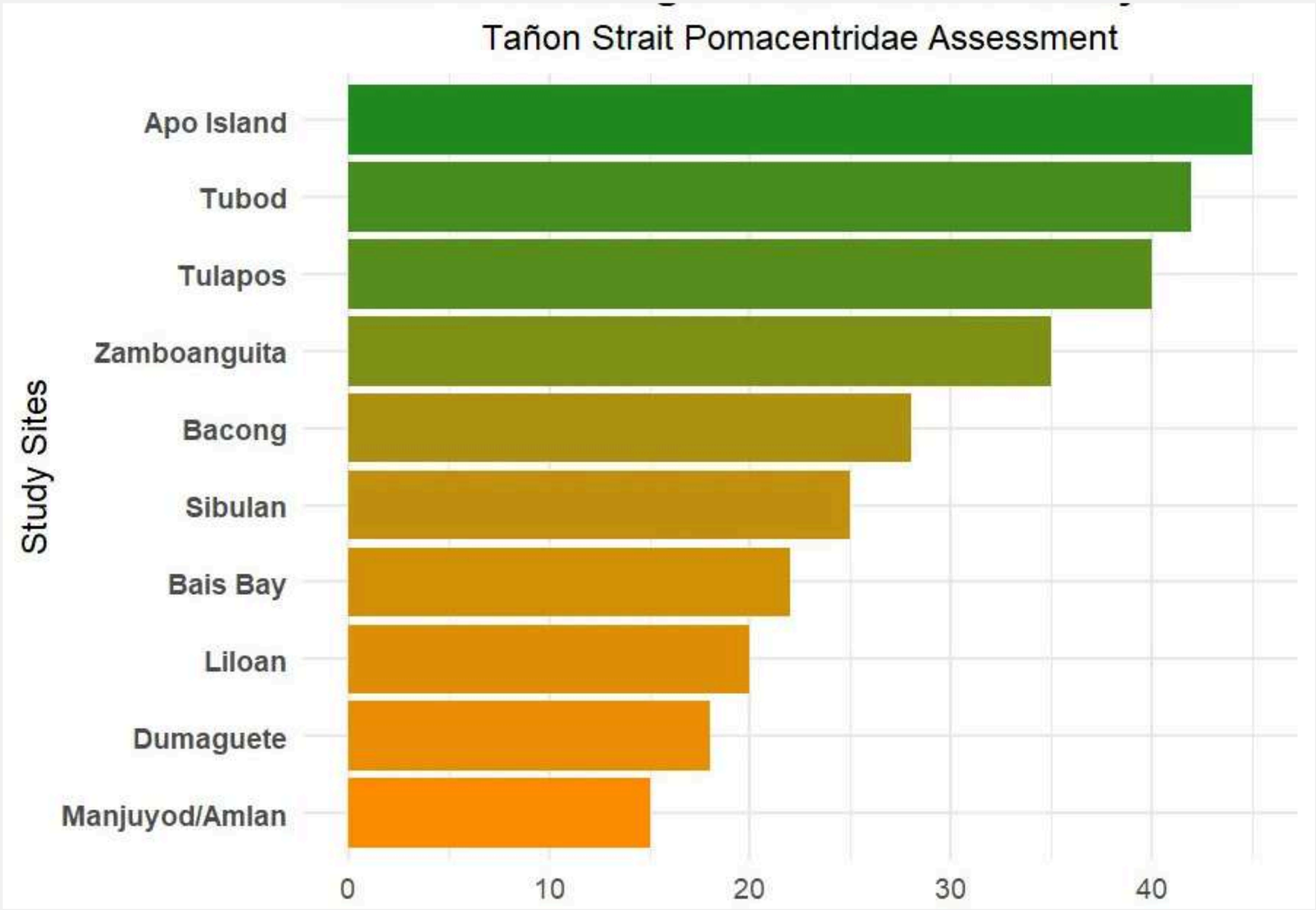
presented by:

Calimbo
Corporal
Descartin
Legwis
Namacpacan
Relon
Zapanta

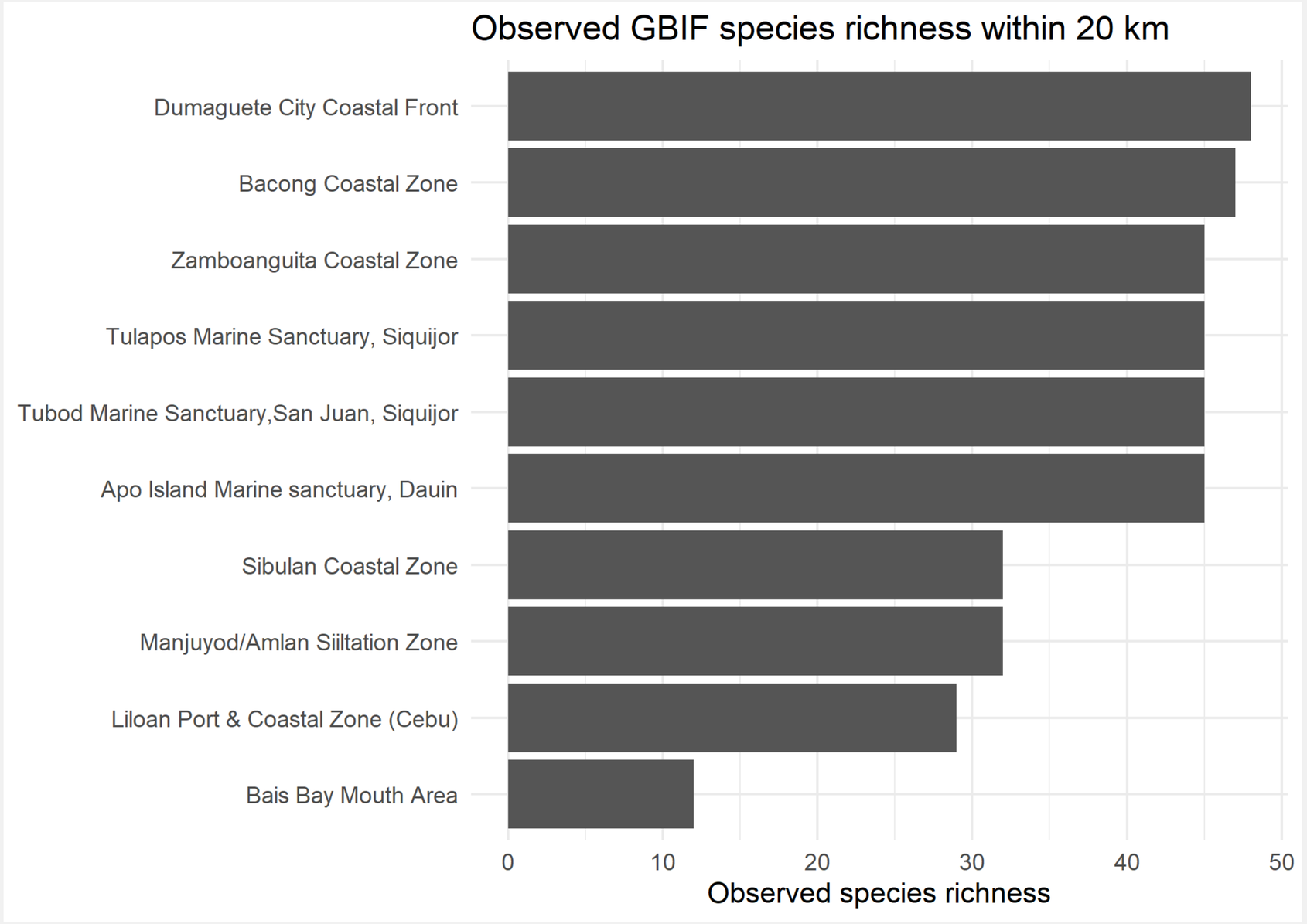
Study Area


Study Sites Across the Tañon Strait Gradient
Pomacentridae Biodiversity Assessment (n = 10 sites)
Bais Bay
Manjuyod/Amlan
Sibulan
Liloan
Tubod Marine
Dumaguete
Bacong
Tulapos
Zamboanguita
Apo Island
9.6
9.5
9.4
9.3
9.2
9.1
Latitude (°N)
123.2
123.3
123.4
Longitude (°E)
Site Category
Coastal
Sanctuary
Silted/Mainland
Urban/Coastal
a
a
a
a

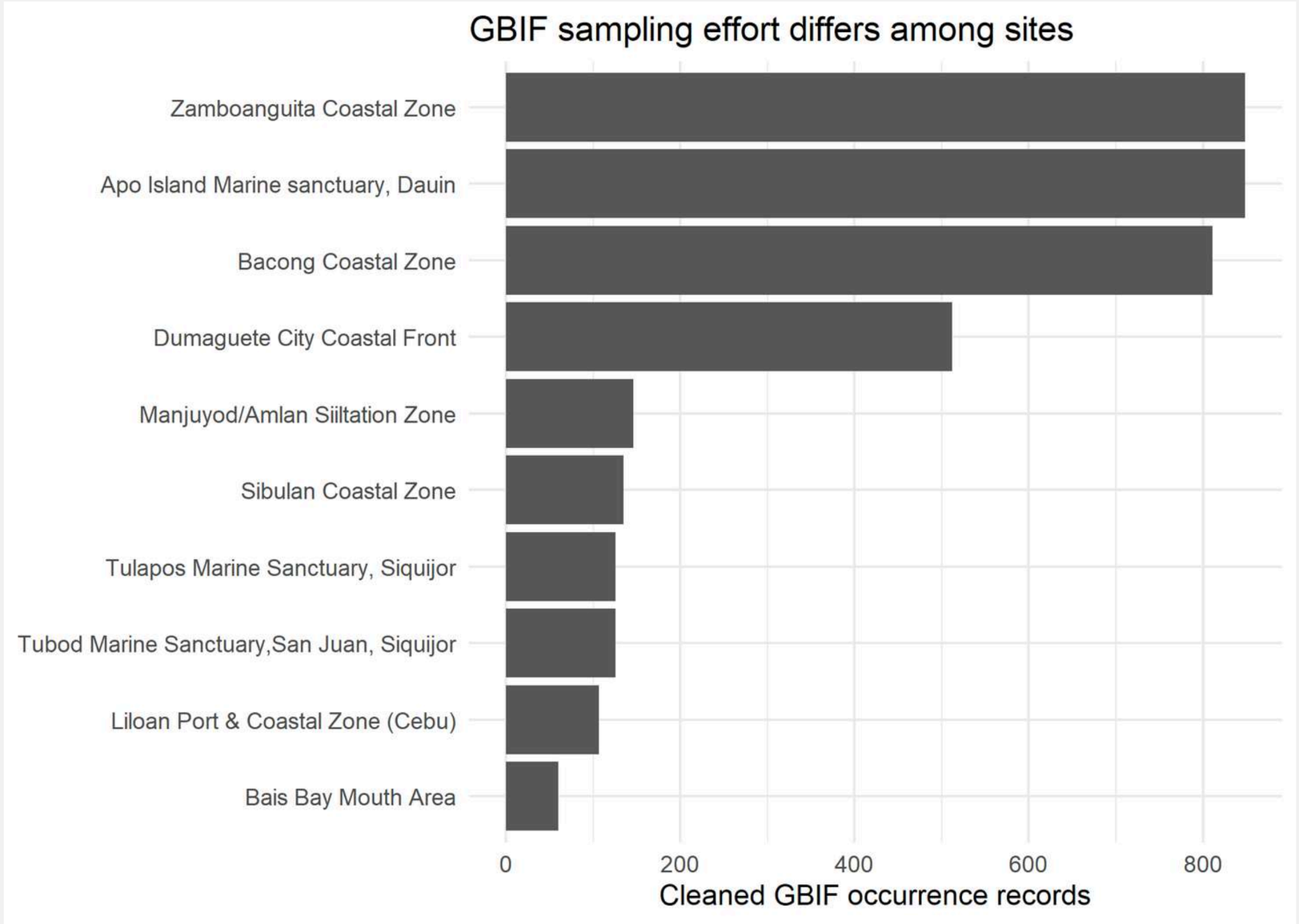
GBIF Data & Spatial Buffers



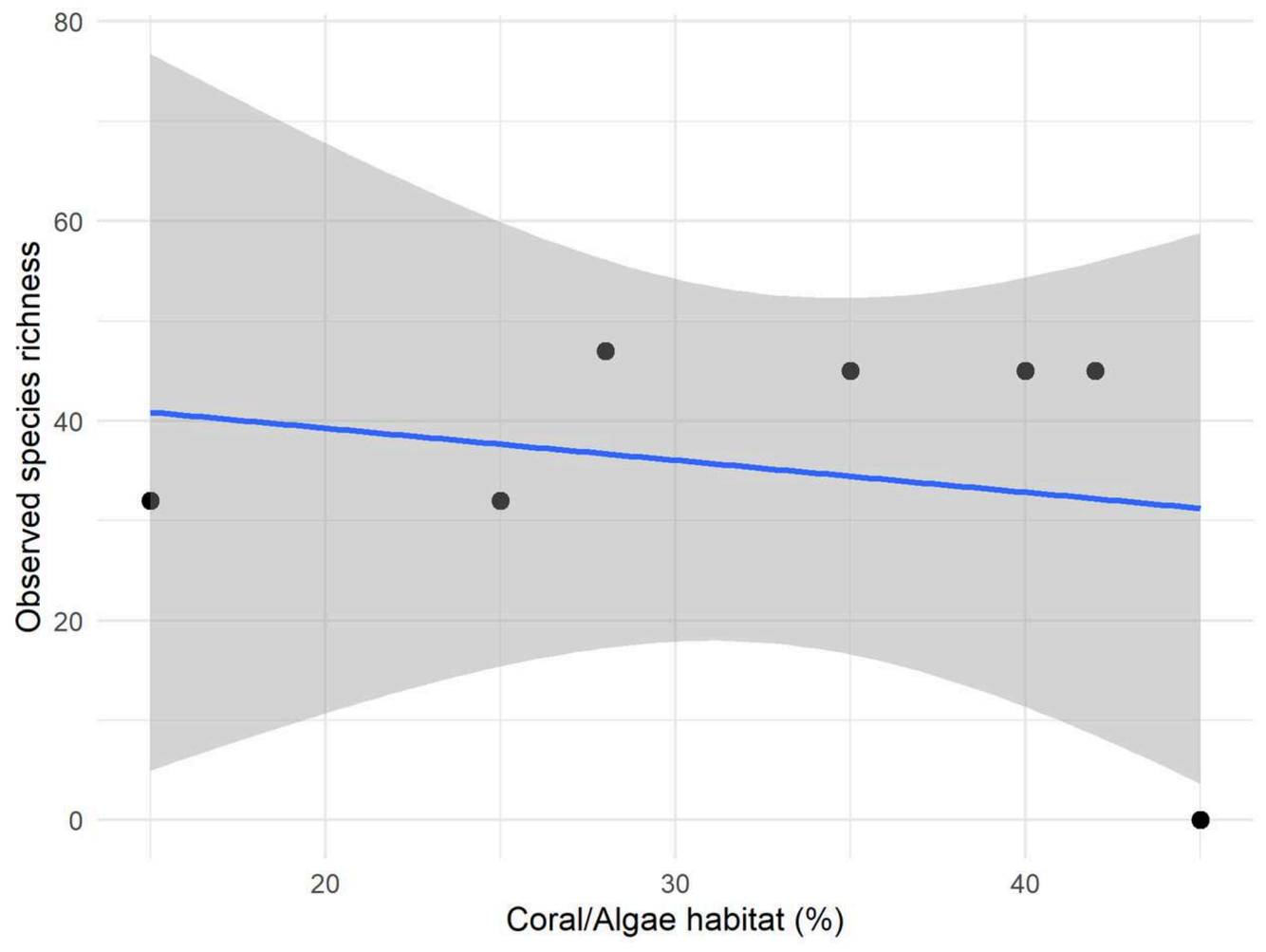
Biodiversity Summary & Richness



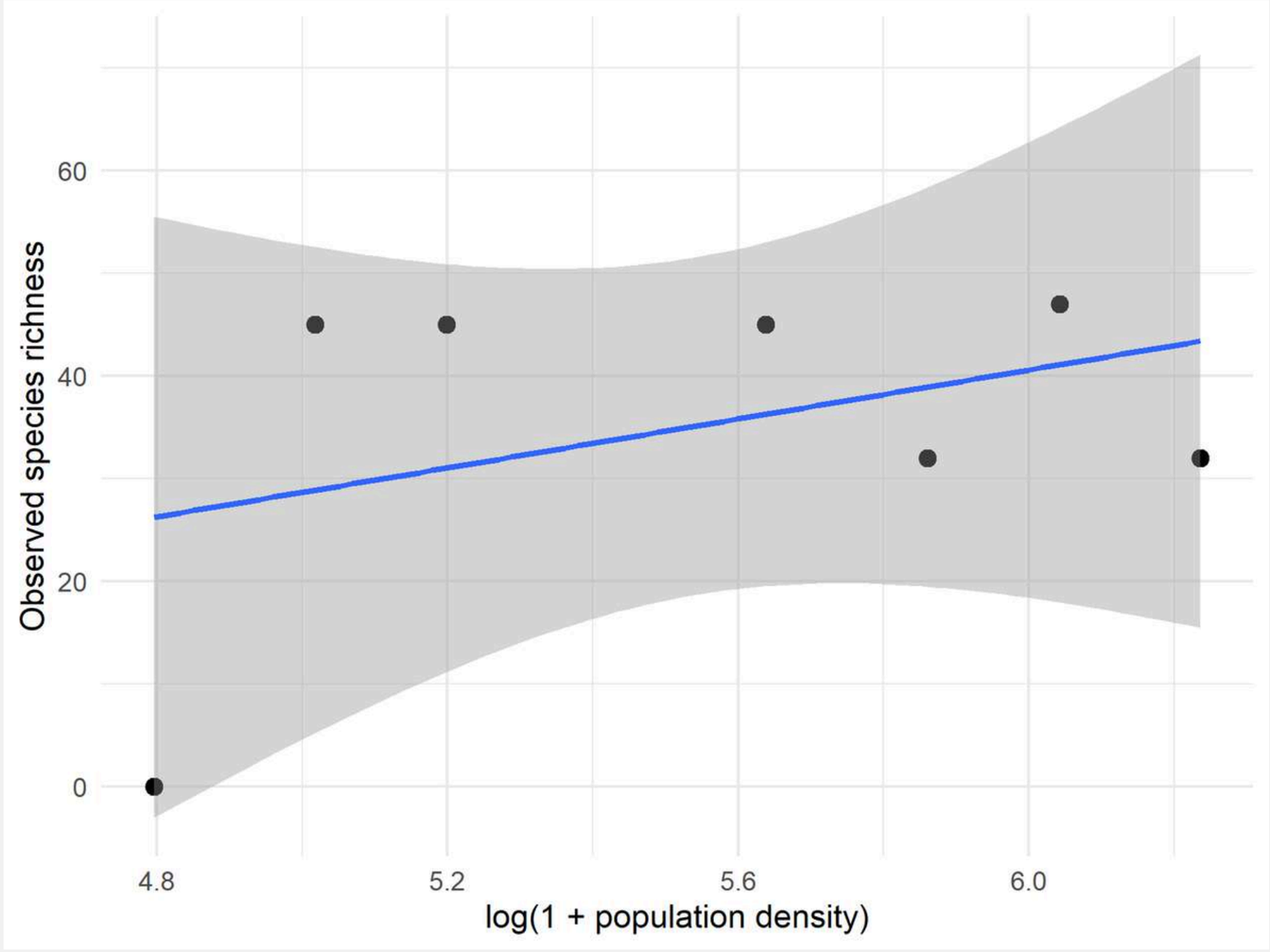
Sampling Effort vs. Richness



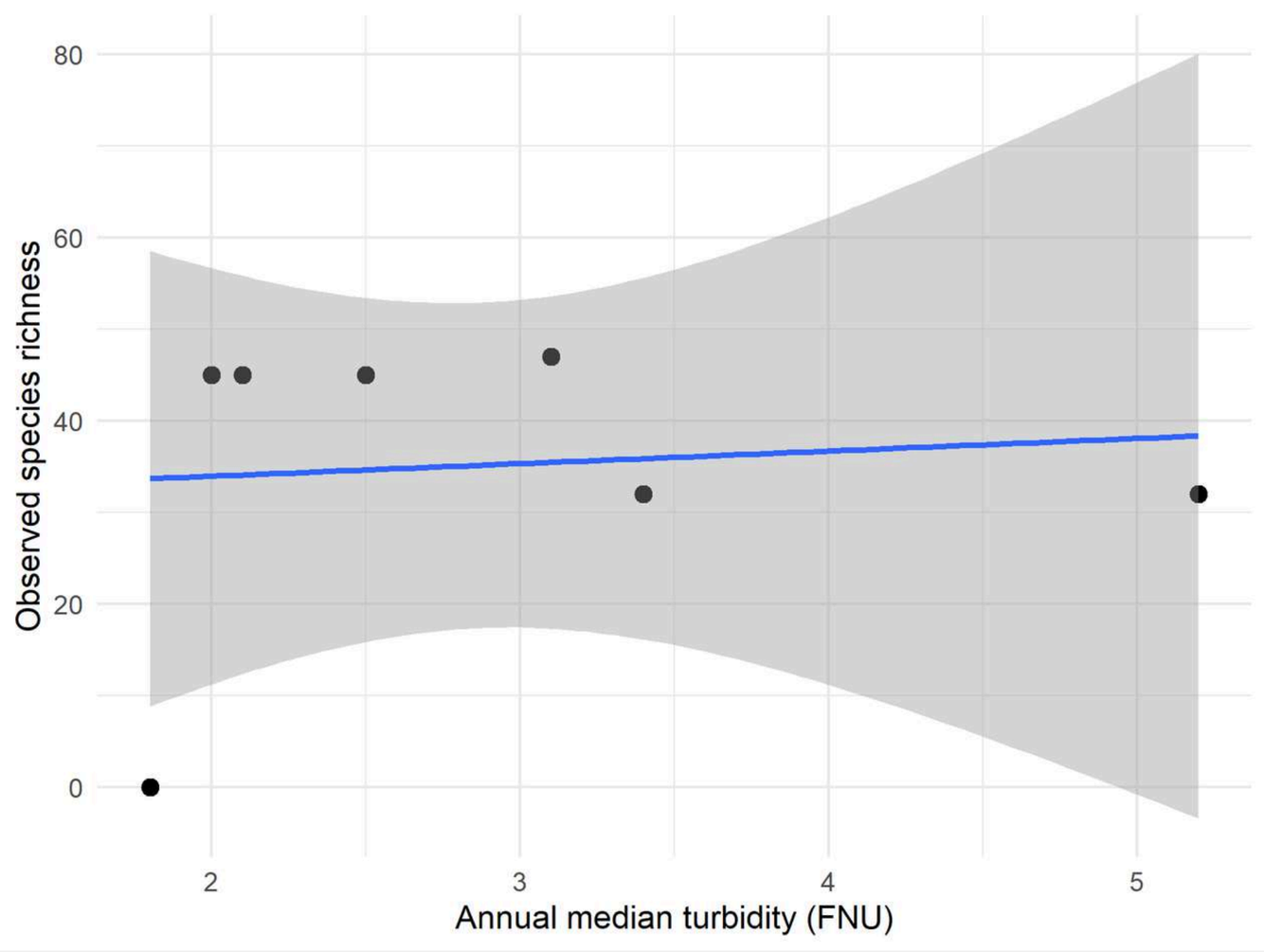
Environment Metrics & Bethnic Composition



Environment Metrics & Bethnic Composition



Environment Metrics & Bethnic Composition



Exploratory GLM Results

```
EXPLORATORY GLM RESULTS
=======================

Candidate A: habitat + turbidity + reef-area offset

Call:
glm(formula = observed_species_richness ~ scale(coral_algae_pct) +
    scale(turbidity_fnu) + offset(log(reef_area_km2)), family = poisson(link = "log"),
    data = model_data)

Coefficients:
                    Estimate Std. Error z value Pr(>|z|)
(Intercept)          2.06968    0.06522  31.736  < 2e-16 ***
scale(coral_algae_pct) -1.22799    0.29304  -4.191 2.78e-05 ***
scale(turbidity_fnu)   -1.27977    0.29313  -4.366 1.27e-05 ***
---
Signif. codes:  0 â€˜***â€™ 0.001 â€˜**â€™ 0.01 â€˜*â€™ 0.05 â€˜.â€™ 0.1 â€˜ â€™ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 95.725  on 6  degrees of freedom
Residual deviance: 75.497  on 4  degrees of freedom
AIC: 114.75

Number of Fisher Scoring iterations: 5


Dispersion statistic A: 13.26016

Candidate B: population + turbidity + reef-area offset

Call:
glm(formula = observed_species_richness ~ scale(log1p(population_density_km2)) +
    scale(turbidity_fnu) + offset(log(reef_area_km2)), family = poisson(link = "log"),
    data = model_data)

Coefficients:
                                 Estimate Std. Error z value Pr(>|z|)
(Intercept)                       2.07118    0.06507  31.832  < 2e-16 ***
scale(log1p(population_density_km2))  0.37676    0.09626   3.914 9.08e-05 ***
scale(turbidity_fnu)                 -0.35062    0.09971  -3.516 0.000438 ***
---
Signif. codes:  0 â€˜***â€™ 0.001 â€˜**â€™ 0.01 â€˜*â€™ 0.05 â€˜.â€™ 0.1 â€˜ â€™ 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 95.725  on 6  degrees of freedom
Residual deviance: 77.980  on 4  degrees of freedom
AIC: 117.23

Number of Fisher Scoring iterations: 5


Dispersion statistic B: 13.6111

Interpretation warning:
These are exploratory models with only about 10 sites and opportunistic GBIF records.
Do not interpret associations as causal effects. Check sampling effort and ecological plausibility.
```



Conclusion & Limitations

- Highest at Dumaguete City Coastal Front (n = 44)„ lowest at Bais Bay (n = 12).
- No clear effect of environment.
- Pattern is driven by sampling bias.
- Buffer too large.
- Uneven GBIF effort.

Thank you





